

Diagonality

The Best of Both Worlds

The tactical blog Spielverlagerung argues the diagonal is football's optimum: the progression of a vertical pass with the safety of a horizontal one.

We put that theory to the test on 3D skeleton data, and turn it into a live map of where to attack.

Jaime Oriol

DIAGONALITY: THE ORIENTATION-AWARE PIPELINE

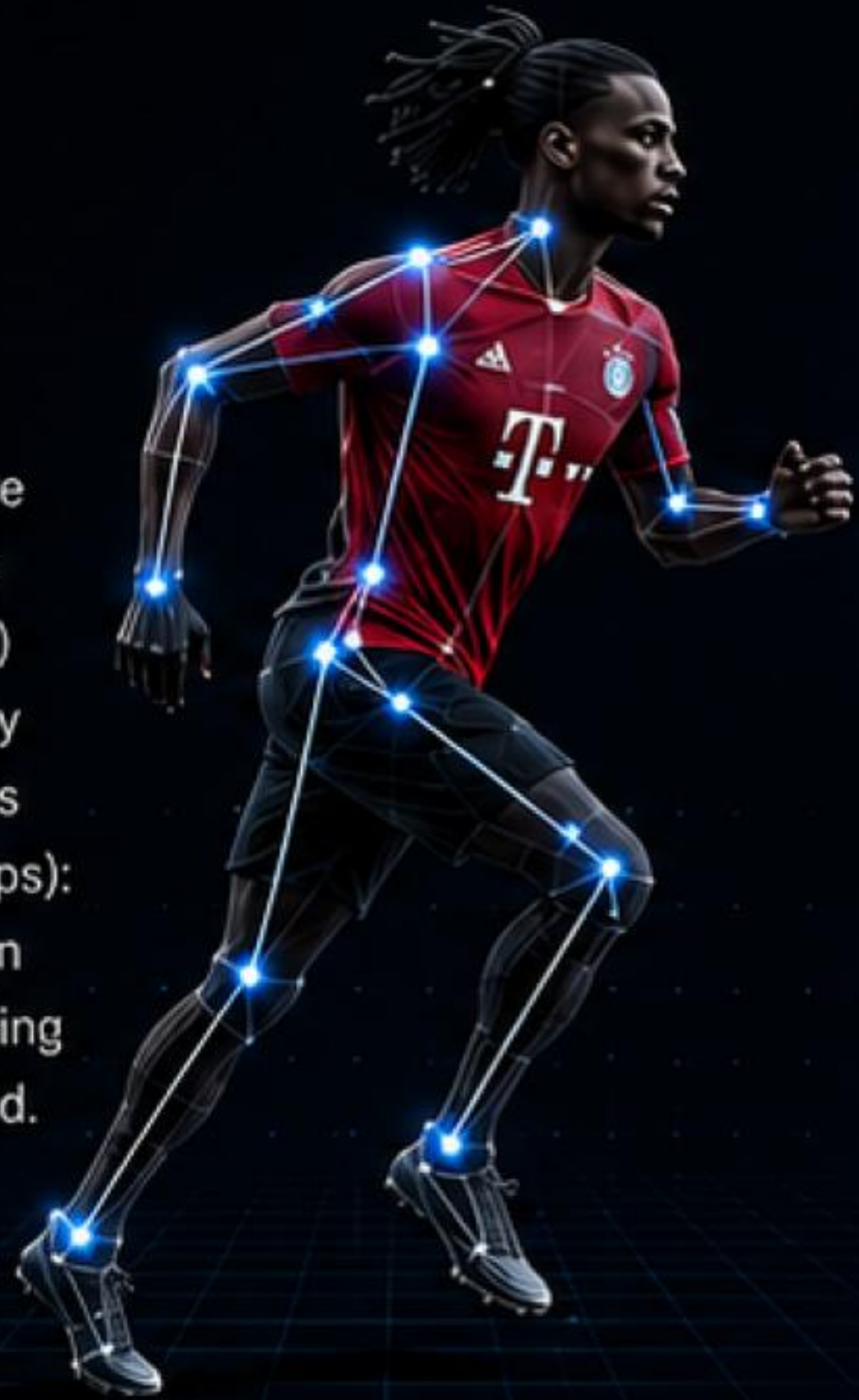
FROM 3D SKELETON TRACKING TO A REAL-TIME MAP OF DIAGONAL OPPORTUNITY

• AWS WORLD SPORTS INNOVATION CUP 2026 • CHALLENGE 2 •

INPUT 1 · 3D SKELETON

21 body
keypoints
at 50 Hz.

We read where
each player is
looking (head)
and which way
the body faces
(shoulders, hips):
the orientation
ordinary tracking
never captured.



MICHAEL OLISSE
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1 · VISION

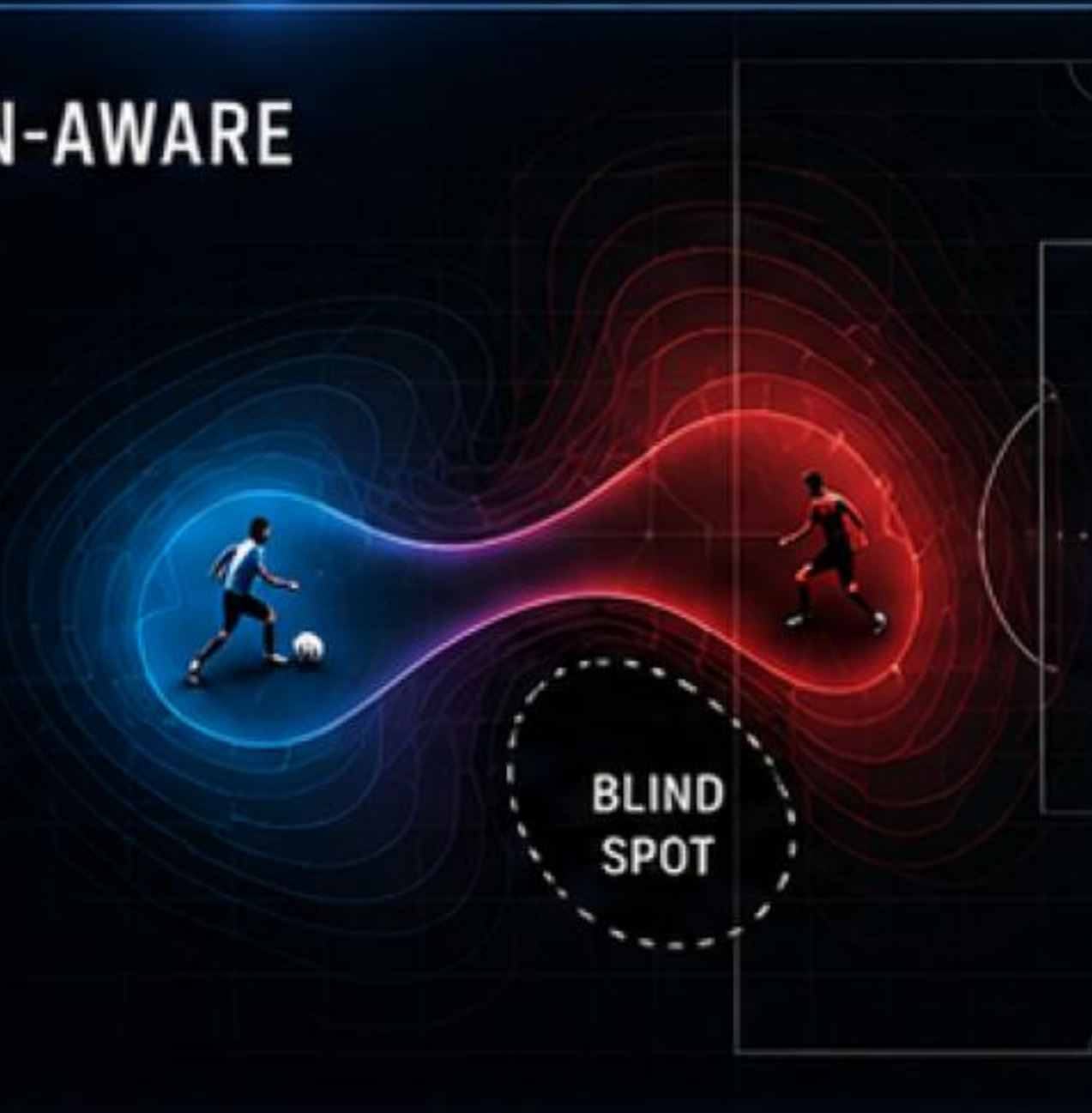
What every defender
can actually see.

A probabilistic 120°
field-of-view cone
around the real head
direction. It narrows
as the player runs
faster, and other
players cast occlusion
shadows.

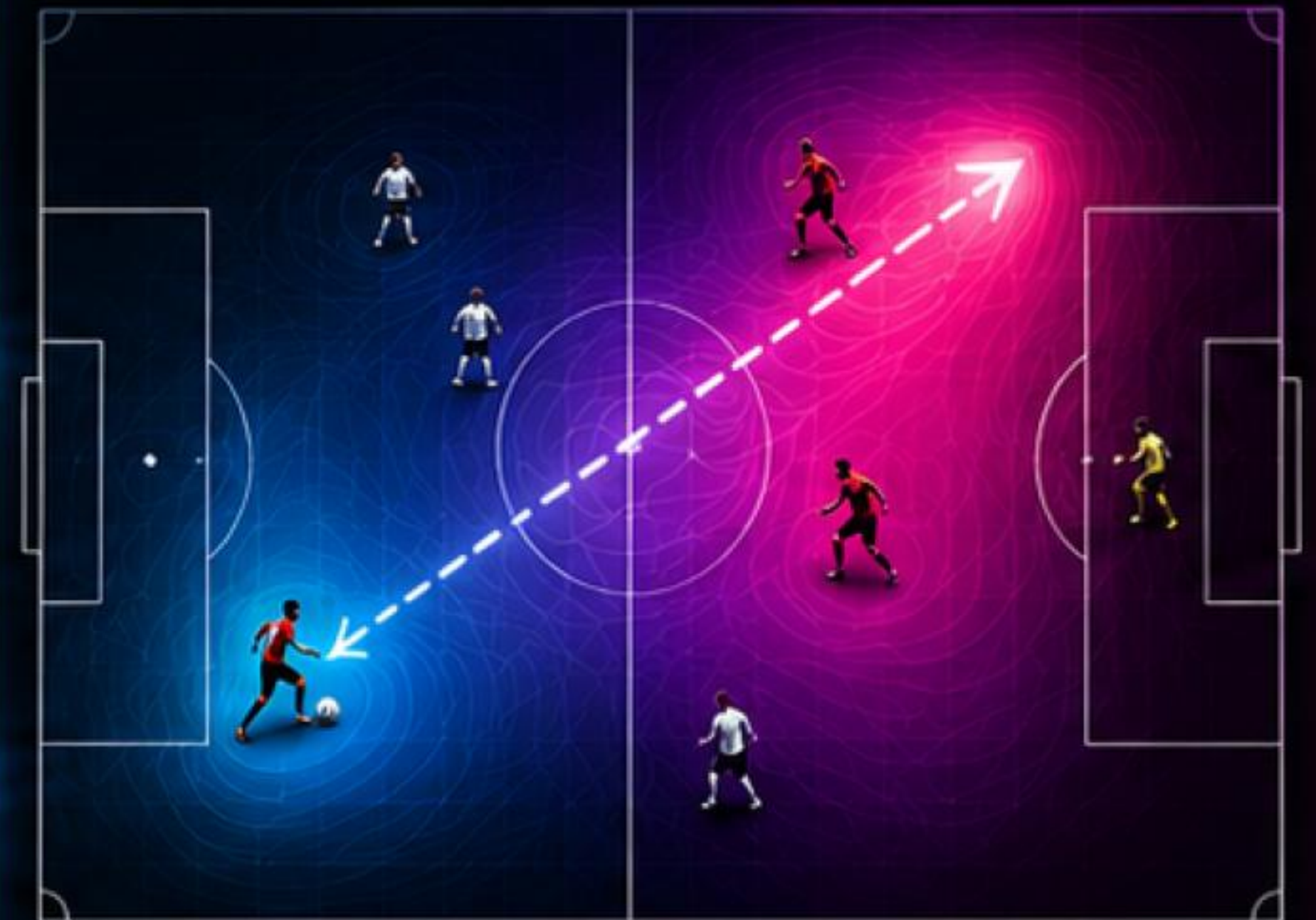


2 · ORIENTATION-AWARE PITCH CONTROL

Not plain pitch control.
What every player can
reach given which way
they are turned: reaching
for a threat behind the
shoulder costs reaction
plus turning time, so the
control blob collapses in
the blind spot.



3 · DOS: DIAGONAL OPPORTUNITY SURFACE



Vision and pitch control fused. The extra control an
attacker buys by delivering the ball diagonally (into
a defender's blind side) instead of straight. A blind
defender reacts late, his reach collapses, and DOS
lights up the gap.

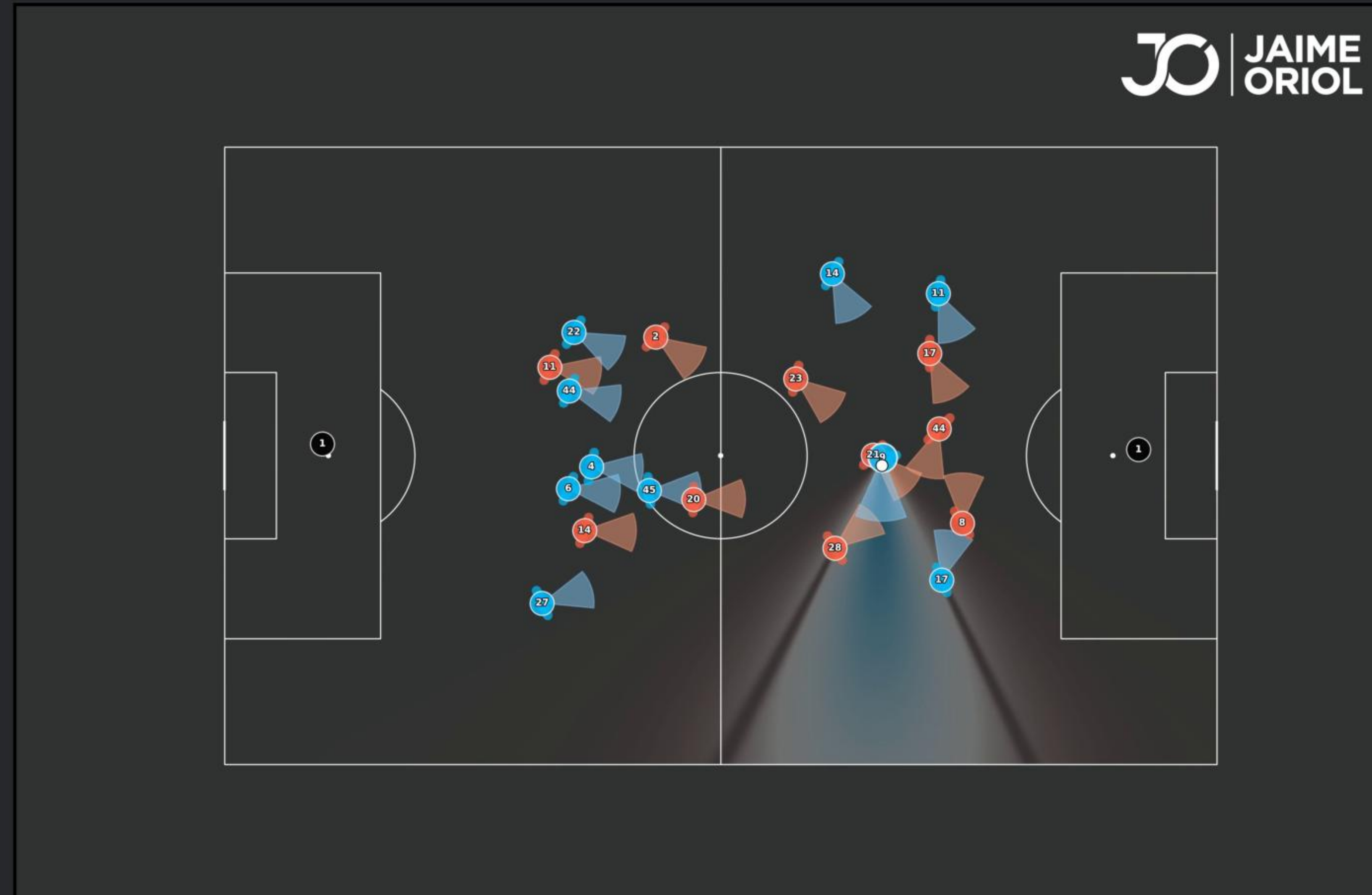


TRACAB 3D SKELETON

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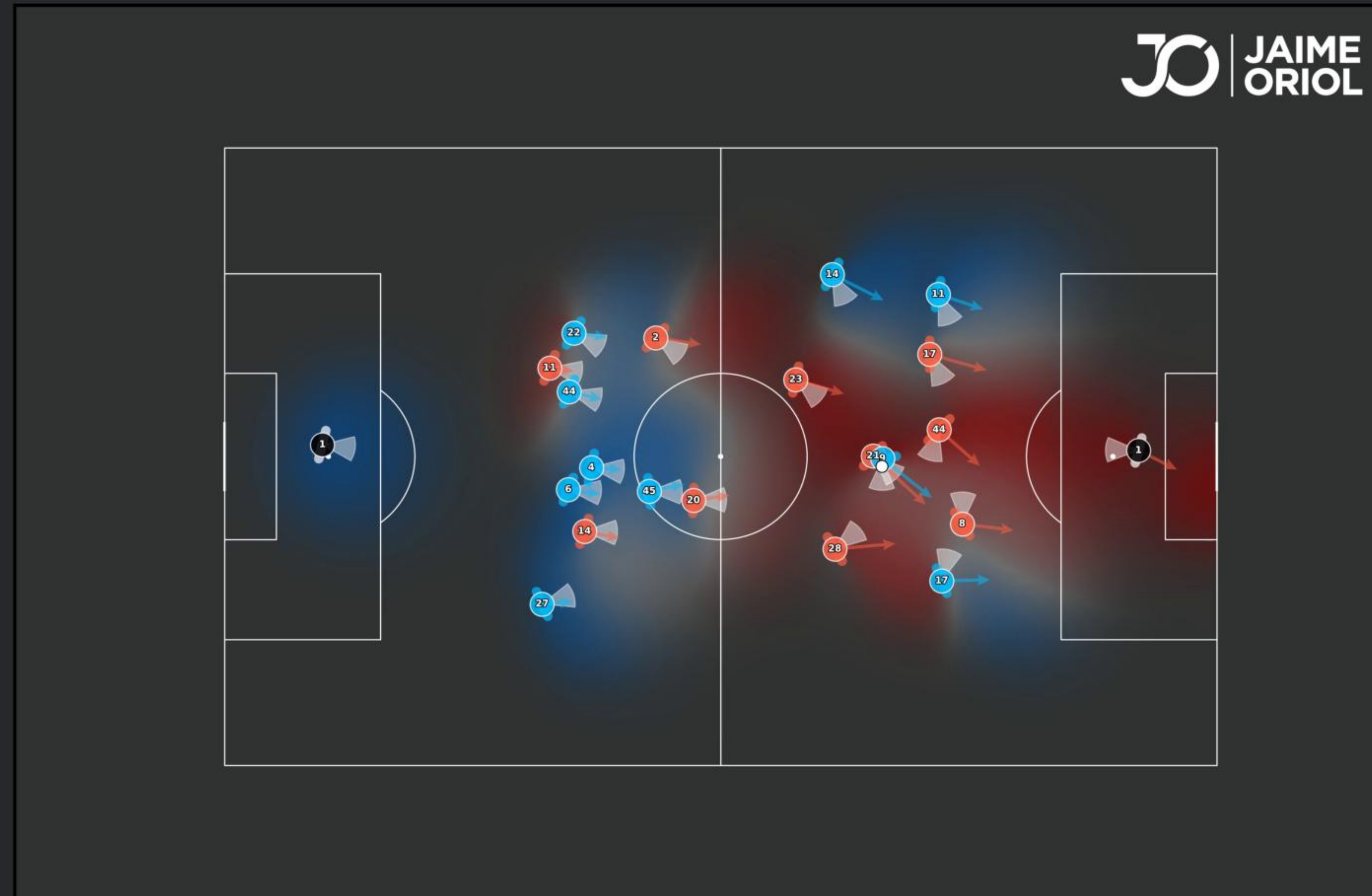
CREATED BY JAIME ORIOL

What every player can see



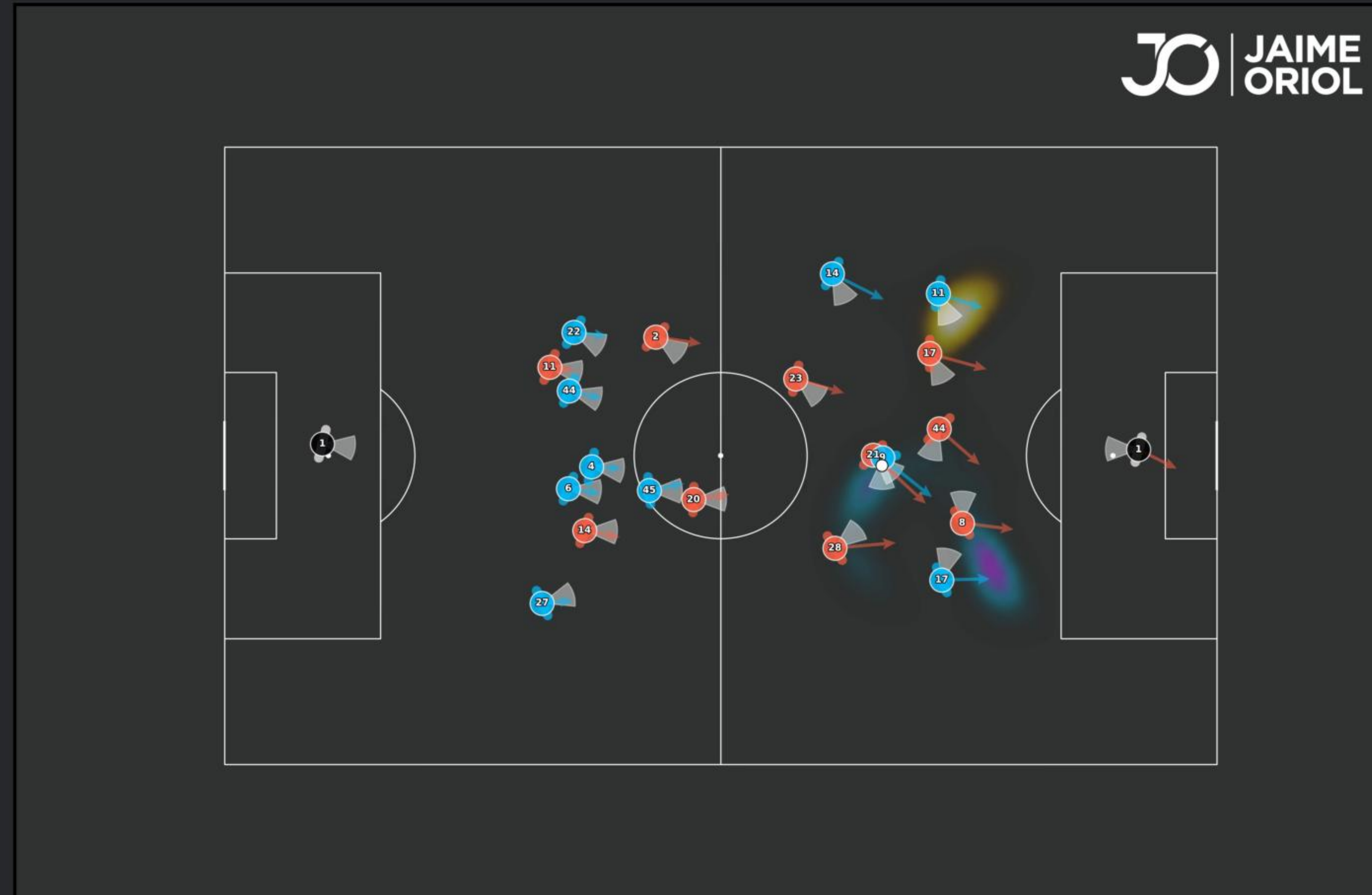
A 120° cone of vision built from where each player is really looking. It narrows when they sprint, and other players block the view.

What every player can reach



How much ground a player covers depends on which way he faces. Behind his shoulder it shrinks, leaving a real gap in the defence.

The Diagonal Opportunity Surface



The extra space a diagonal pass opens up, shown only where the player on the ball can actually see it. Cyan: he sees it. Amber: he lost track of it.